

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – IV Examination – APRIL 2012

PT208 BIOSTATISTICS

Date: 1.05.12, Tuesday

Time: 10.00a.m to 10.10 a.m

Maximum Marks: 05

Instructions:

1. There are two parts in the paper. **PART – I** contains 5 Questions and **PART – II** contains 2 Questions.
2. **PART – I** should be answered in MCQ'S question paper & **PART – II** should be answered in answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 10 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 10 min, can collect main question paper for **PART – II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs)

(5 × 1 = 5)

(Encircle the correct answer)

1. Variance is _____ of the Standard deviation.
 - a. Half.
 - b. Square.
 - c. +ve square root.
 - d. None of the above.
2. Data given of Systolic blood pressure is an example of _____.
 - a. Qualitative Data
 - b. Quantitative Data
 - c. None of (a) and (b)
 - d. Both of (a) and (b)
3. How many parameters does Poisson distribution have?
 - a. 0
 - b. 1
 - c. 2
 - d. 3
4. Data of the diabetic patients taken from any monthly magazine will be considered as _____.
 - a. Secondary data
 - b. Primary data
 - c. Both of (a) and (b)
 - d. None of (a) and (b)
5. Pie chart can be used for _____.
 - a. Categorical data
 - b. Continuous data
 - c. Both of (a) and (b)
 - d. None of (a) and (b)

CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY CHARUSAT CAMPUS CHANGA	
Examination Semester _____	
Subject _____ No. _____	
Candidate's Seat No. _____	
Section _____	
Date _____	
Sr. Supervisor's Sign.	

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – IV Examination – APRIL 2012

PT208 BIOSTATISTICS

Date: 1.05.12, Tuesday

Time: 10.10a.m to 11.30 a.m

Maximum Marks: 30

Instructions:

1. There are two parts in the paper. PART –I contains 5 Questions and PART – II contains 2 Questions.
2. PART –I should be answered in MCQ'S question paper & PART – II should be answered in answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for PART – II is 1 Hr 20min.
5. Draw the figure wherever necessary.

PART-II

Q.1 Answer any one of the following:

(1 × 10 = 10)

- (a) The amount of chemical compound Y, which were dissolved in 100 grams of water at various temperatures, X were recorded as follows

X (°C)	15	15	30	30	45	45	60	60
Y(grams)	12	10	25	21	31	33	44	39

Find the line of regression of y on x and estimate the amount of chemical that will dissolve in 100 grams of water at 55°C.

- (b) Data given below table consists of Systolic blood Pressure (in mm Hg) of 30 patients.

130	140	110	115	140	128	124	120	142	110
106	120	130	156	122	123	138	138	134	120
124	156	114	122	108	110	120	138	132	106

- (i) Draw Histogram and give your conclusion.(Use Class Interval = 10)
- (ii) Obtain A. M. for grouped data.

Q.2 Answer any four of the following:

(4 × 5 = 20)

- (a) Calculate mean, median and mode for the 13 observations given below.
7, 2, 3, 1, 9, 5, 7, 6, 5, 7, 1, 9 and 7.
- (b) Find D_5 and P_{75} using the raw data of Systolic blood Pressure given in Q.1 (b).
- (c) A box contains 10 tablets of Aspirin, 9 tablets of Nurofen and 5 tablets of Acetaminophen. If one tablet is chosen at random from the box, find the probability that (i) It is Aspirin (ii) It is not Aspirin.
- (d) Explain following terms with relevant Example
(i) Discrete Variable (ii) Continuous Variable (iii) Population
(iv) Sample (v) Parameter
- (e) The following table shows the results of an experiment to investigate the effect of vaccination induced on the animals against a particular disease. Use chi-square test to test the hypothesis that there is no difference between the vaccinated and unvaccinated groups i.e. vaccination and this disease are independent.

$$\chi^2(1, 0.05) = 3.84$$

	Got disease	Did not got disease
Vaccinated	9	42
Not vaccinated	17	28